



Suppression of miR426 enhances photoprotection and chlorophyll stability via upregulation of *ELIP2* in *Arabidopsis thaliana*

Arif Faisal Mustaffa¹ · Syed Muhammad Iqbal Syed Othman¹ · Nur Irdina Izzatie Mohd Zahid¹ · M. Hafiz Che-Othman¹ · Hoe-Han Goh² · Muhammad Sajad³ · Ismanizan Ismail^{1,2,4}

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Abstract

Background Plants require light energy to undergo photosynthesis, a process that is essential for their growth and development. However, excessive light can lead to photooxidative stress and photoinhibition, compromising key photosynthetic components, particularly chlorophyll pigments. Early light-induced protein 2 (ELIP2) plays a key role in photoprotection by suppressing chlorophyll biosynthesis, preventing chlorophyll bleaching, and enhancing photosynthetic efficiency. The expression of *ELIP2* is regulated through multiple mechanisms, including post-transcriptional and translational controls. This study aims to elucidate the role of plant microRNA (miRNA) in regulating *ELIP2* gene activity at the post-transcriptional level, thereby enhancing plant defence mechanisms and maintaining chlorophyll stability under conditions of photooxidative stress and photoinhibition.

Methods and results An in silico analysis identified miR426 as a key post-transcriptional regulator of *ELIP2*. Using short tandem target mimic (STTM), transgenic *Arabidopsis thaliana* plants constitutively expressing STTM426 were developed and exposed to varying light intensities alongside wild type (WT) plants. Expression profiling revealed a significant reduction in miR426 levels and increased *ELIP2* transcript abundance in transgenic plants compared to WT ($n=3$). Interestingly, transgenic plants displayed minimal changes in chlorophyll content under normal light and maintained stability even under extreme light conditions ($n=9$). Gas exchange and chlorophyll *a* fluorescence analyses further demonstrated enhanced photosynthetic efficiency and photoprotective mechanisms in the transgenic plants ($n=9$). These physiological improvements were supported by morphological observations, as transgenic plants consistently exhibited reduced rosette diameter and total biomass across all light intensities ($n=9$).

Conclusion These preliminary findings suggest that suppression of miR426 may serve as a promising strategy to enhance plant resilience against photooxidative and photoinhibitory stress by strengthening photoprotective mechanisms and maintaining chlorophyll stability.

Keywords Abiotic stress · Early light-induced protein 2 (ELIP2) · High light stress · MicroRNA (miRNA) regulation · Post-transcription regulation · Short tandem target mimic (STTM)

✉ Ismanizan Ismail
maniz@ukm.edu.my

¹ Department of Biological Sciences and Biotechnology, Faculty of Science and Technology, Universiti Kebangsaan Malaysia (UKM), Bangi 43600, Selangor, Malaysia

² Institute of Systems Biology, Universiti Kebangsaan Malaysia (UKM), Bangi 43600, Selangor, Malaysia

³ Department of Plant Breeding and Genetics, Faculty of Agriculture and Environment, The Islamia University of Bahawalpur (IUB), Bahawalpur 63100, Punjab, Pakistan

⁴ GENIUS@Pintar National Centre, Universiti Kebangsaan Malaysia (UKM), Bangi 43600, Selangor, Malaysia

Abbreviations

A	CO ₂ Assimilation Rate
cDNA	Complementary DNA
E	Transpiration Rate
ELIP2	Early Light-Induced Protein 2
F_v'/F_m'	Maximum Quantum Efficiency of PSII Under Light-Adapted Conditions
gDNA	Genomic DNA
Gsw	Stomatal Conductance Rate
miR426	MicroRNA426
miRNA	MicroRNA
mRNA	Messenger RNA

NCBI	National Center for Biotechnology Information
NPQ	Non-Photochemical Quenching
PPFD	Photosynthetic Photon Flux Density
PSII	Photosystem II
qP	Photochemical Quenching of Chlorophyll Fluorescence
qRT-PCR	Quantitative Real-Time PCR
ROS	Reactive Oxygen Species
STTM	Short Tandem Target Mimic
STTM426	Short Tandem Target Mimic for miR426
TAIR	The Arabidopsis Information Resource
TBP	Tetrapyrrole Biosynthesis Pathway
WT	Wild Type

Introduction

Abiotic stress is a major factor influencing plant molecular, physiological, and morphological characteristics, significantly impacting their productivity [1–3]. One of the most common abiotic stresses plants faces is fluctuations in light intensity. Light energy is essential for photosynthesis, driving plant growth and development. However, excessive light can cause oxidative damage, disrupting the photosynthetic process [4]. High light intensity promotes the formation of reactive oxygen species (ROS), leading to photooxidative stress [5]–[6]. Although ROS plays a signalling role in activating defence mechanisms at moderate levels, excessive ROS accumulation leads to severe photodamage. This results in photoinhibition, where chlorophyll pigments are damaged, eventually reducing the plant's photosynthetic efficiency [7].

As a protective response to high light stress and to maintain normal physiological function, plants activate early light-inducible proteins (ELIPs). These proteins safeguard chlorophyll from degradation and enhance adaptation to extreme light conditions [8]. The *ELIP* gene has been studied across various plant species, including *Arabidopsis thaliana*, *Oryza sativa* (rice), *Pisum sativum* (pea), *Hordeum vulgare* (barley), *Nicotiana tabacum* (tobacco), *Brassica oleracea* (cabbage), *Solanum lycopersicum* (tomato), and *Populus alba* (poplar) [9–16]. While several photoprotective genes such as *PHOTOSYSTEM II SUBUNIT S* (PsbS) [17], *ONE-HELIX PROTEINS (OHP1/2)* [18]–[19], and *STRESS-ENHANCED PROTEINS (SEP1/2)* [20] contribute to light stress resilience, *ELIP* particularly *ELIP2* stands out as uniquely pivotal. Unlike these functionally specialized proteins, *ELIP2* integrates light stress signalling with metabolic regulation of chlorophyll biosynthesis via the tetrapyrrole biosynthesis pathway (TBP). This dual functionality positions *ELIP2* not only as a molecular safeguard against

photooxidative damage but also as a central regulatory node within the broader photoprotective network, thereby justifying its selection as the primary focus of this study.

In higher plants, *ELIP2* serves as a key component in responding to photooxidative stress and photoinhibition by minimizing chlorophyll degradation and maintaining photosynthetic stability under excessive light exposure [8]. *ELIP2* has been reported to specifically protect photosystem II (PSII) under high light stress through its interaction with the TBP pathway, functioning by reducing the accumulation of free chlorophyll and preventing photooxidative damage [13]–[14]. This regulation of chlorophyll biosynthesis is achieved by modulating the activity of key enzymes within the TBP pathway, thereby preventing chlorophyll degradation and enhancing plant resilience to high light stress [21]. However, under excessive light exposure, *ELIP2* production may be insufficient to fully protect the plant, resulting in chlorophyll degradation due to photooxidative stress and photoinhibition. The gradual accumulation of *ELIP2* suppresses the TBP pathway responsible for free chlorophyll synthesis, particularly when photosystem II (PSII) damage increases by up to 40%, under extreme light intensities ranging from 1,500 to 2,000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ [22].

Furthermore, *ELIP2* gene expression is regulated through multiple mechanisms, including post-transcriptional and translational control, to precisely modulate *ELIP2* protein levels, with plant microRNAs (miRNAs) playing a key role in this process. Plant miRNAs are small, non-coding RNA molecules (20–24 nucleotides) encoded in the nucleus that regulate gene expression at the post-transcriptional level through mRNA degradation and translational repression [23]. Implementing an in silico approach, this study proposes that miR426-mediated post-transcriptional regulation reduces plant resilience to photooxidative stress and photoinhibition by targeting *ELIP2* mRNA for degradation or translational repression. Although there are limited previous studies, miR426 have been shown to play a significant role in the arsenic stress response of *Brassica juncea* [24]. Additionally, miR426 has been identified as a regulator of probable *TREHALOSE-PHOSPHATE PHOSPHATASE 2 (TPP2)* in *Helianthus annuus*, where it influences plant growth and exhibits distinct expression patterns under various abiotic stresses, including heat, drought, salt, and cadmium stress [25]. These findings suggest that miR426 contributes to plant stress adaptation by modulating key metabolic pathways. Despite the recognised importance of *ELIP2*, its post-transcriptional regulation remains unclear. This study investigates whether miR426 plays a regulatory role in *ELIP2* expression under conditions of photooxidative stress and photoinhibition.

To better understand the role of miRNAs in *ELIP2* regulation, this study investigates how the downregulation

of miR426 influences chlorophyll homeostasis and plant defence mechanisms under photooxidative stress. While various methods have been developed to suppress miRNAs, this study utilizes short tandem target mimic (STTM) technology as an endogenous loss-of-function (LoF) strategy by generating transgenic *A. thaliana* plants carrying *STTM426* constructs. This approach enables targeted suppression of miR426, allowing for a direct evaluation of its regulatory effect on *ELIP2*. We further analysed expression profiles by quantifying the relative abundance of miR426 and *ELIP2* transcripts, enabling a direct assessment of miR426-*ELIP2* interactions and the impact of STTM application. Additionally, physiological and morphological traits including chlorophyll content, photosynthetic efficiency, photoprotective mechanisms, rosette diameter, and total biomass, are analysed and compared between *STTM426* and wild type (WT) plants. This preliminary study demonstrates the development of plants with enhanced tolerance to photooxidative stress and photoinhibition, where reinforced defence mechanisms and stabilized chlorophyll content contribute to improved photosynthetic efficiency, even under extreme environmental conditions.

Materials and methods

In silico analysis for identification and validation of miRNA targeting *ELIP2*

This study initially employed an in silico approach to identify miRNAs targeting the *ELIP2* transcript, a crucial step for predicting miRNA-mRNA interactions prior to laboratory validation. Algorithms and computational models were utilized to efficiently predict potential miRNA binding sites within the target transcript. The coding sequence of *ELIP2* was retrieved in FASTA format from The Arabidopsis Information Resource (TAIR) database (<https://www.arabidopsis.org/>) and served as the input for subsequent bioinformatics analyses. To ensure sequence accuracy, cross-validation was conducted using the National Centre for Biotechnology Information (NCBI) database (<https://www.ncbi.nlm.nih.gov/>).

Potential miRNAs targeting *ELIP2* were identified using psRNATarget V2 (<https://www.zhaolab.org/psRNATarget/>), a computational tool designed to predict miRNA-mRNA interactions. This tool applies an advanced algorithm to determine miRNA binding sites and their respective heteroduplexes structures [26]. The FASTA sequence retrieved from TAIR was input into psRNATarget V2 using user-defined parameters: Expectation = 5, seed region = 2–13 nt, high-scoring segment pair (HSP) size = 19, gap open penalty = 2, and gap extension penalty = 0.5. The top-ranking

mature miRNA was selected and further validated using the miRBase database (<http://www.mirbase.org/>) [27], which provided detailed insights into its regulatory role in *ELIP2* transcript modulation in *A. thaliana*.

Construction of the miR426 silencing plasmid

The STTM construct used in this study was designed following the method introduced by Tang et al. (2012) [28] and Teotia and Tang (2017) [29], integrating the PmiREN2.0 database (<https://pmiren.com>) [30]. The *STTM426* construct, approximately 146 bp in length, was designed to mimic the complementary binding site of mature miR426, thereby effectively suppressing its activity. A tri-nucleotide bulge (CTA) was introduced into the middle of the *STTM426* sequence to prevent degradation of the complementary binding site. Additionally, two complementary miR426 binding sites were separated by a 48-nt stem-loop spacer, with *SwaI* restriction sites embedded in the middle of each spacer for further modifications. To ensure the efficacy and stability of the construct, additional genetic elements were incorporated, including *HindIII* restriction sites and the CaMV 2 × 35 S (d35S) promoter at the 5' end, along with *EcoRI* restriction sites and the 35 S terminator (T-35 S) at the 3' end, as shown in Supplementary Fig. S1.

The construct was generated by modifying the pOT2-Poly-Cis plasmid, in which the Poly-Cis fragment was replaced with the *STTM426* sequence to produce the pOT2-STTM426 vector. The construction process began with inverse PCR to amplify the *STTM426* fragment using the following primers: Forward primer 5'-GCCATTTAAATATGGTCTAAAGAAGAAGAATCGTAAGGACAACATAATTCCAAAAGAATTCGGTACGCTGAAATCACCAG-3' and reverse primer 5'-GCCATTTAAATTAGACCATAACAACAACAACCTTTTGGAATTAGTTGTCTTACGAAGCTTGGGCTGTCCTCTCCAAATG-3'. These primers incorporated *SwaI* restriction enzyme recognition sites for subsequent cloning into the pOT2-Poly-Cis vector. After successful insertion, self-ligation using *SwaI* restriction enzyme (New England Biolabs) ensured accurate fragment assembly and construct stability. The final pOT2-STTM426 vector was transformed into the competent *Escherichia coli* strain Stellar™ for propagation and validation.

Following validation on chloramphenicol (25 µg/mL), the complete construct including the CaMV 2 × 35 S promoter (d35S), 35 S terminator (T-35 S), and chloramphenicol resistance (CAM^R) selection marker was subcloned into the binary vector pCambia1300-PacI before transformation into *Agrobacterium tumefaciens* (Fig. 1). A *PacI* restriction site was introduced into pOT2-STTM426 via inverse PCR to facilitate ligation, replacing the original *pBR322* origin of replication sequence. This was accomplished using

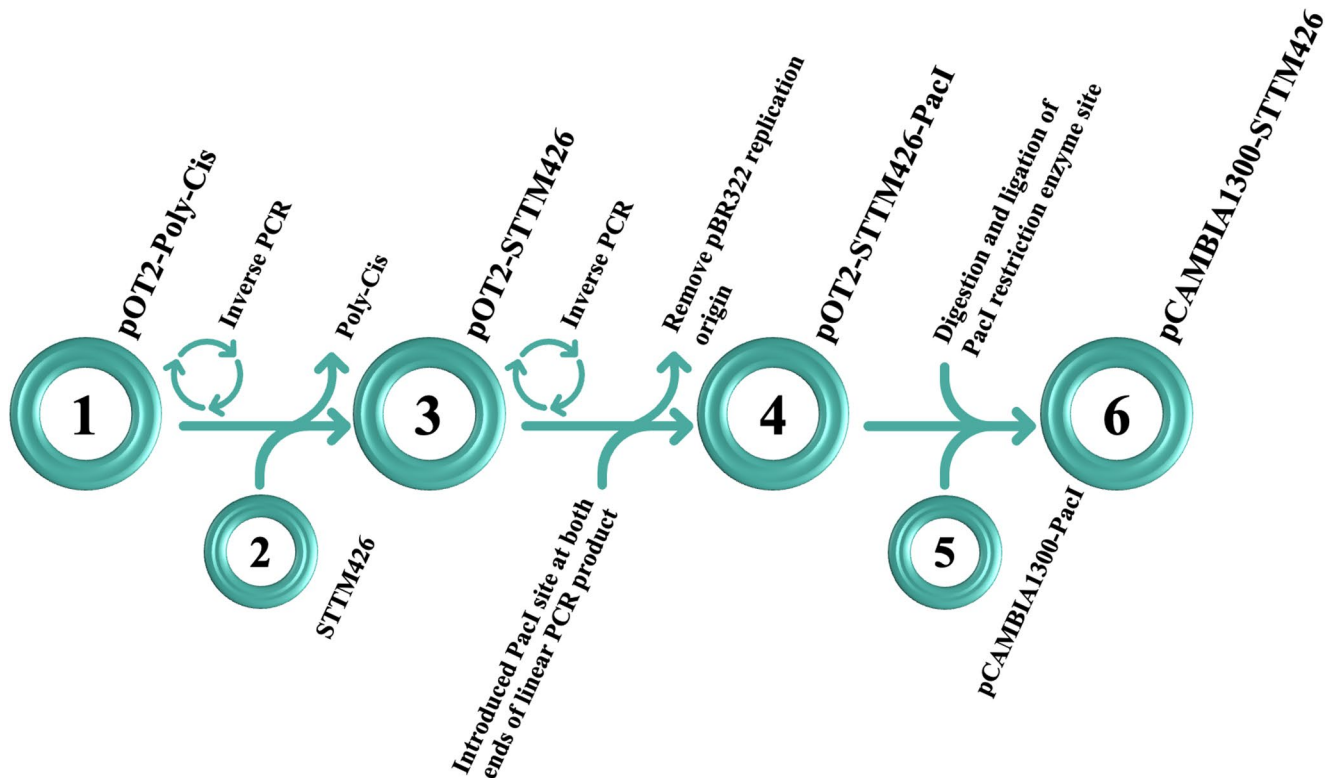


Fig. 1 Schematic flowchart illustrating the development of the constitutive expression construct STTM426

the following primers: Forward primer 5'-TCCCTTAATT AAGTTTGCAAGCAGCAGATTACGCG-3' and reverse primer 5'-TCCCTTAATTAAGAAAGGCGGACAGGTA TCCGGTAAG-3'. The modified vector, pOT2-STTM426-PacI, carrying the *STTM426* insert, was subsequently integrated into pCAMBIA1300-PacI through PacI restriction enzyme (New England Biolabs) and ligated using T4 DNA ligase (Promega), resulting in the recombinant plasmid pCAMBIA1300-STTM426. This final construct contained the *STTM426* expression cassette along with dual antibiotic resistance markers (CAM^R and Kan^R). The pCAMBIA1300-STTM426 plasmid was then transformed into competent *E. coli* StellarTM cells for propagation and validated on the selection plates containing both chloramphenicol (25 µg/mL) and kanamycin (50 µg/mL).

Plant transformation by *Agrobacterium tumefaciens*

The pCAMBIA1300-STTM426 plasmid was transformed into the competent cell of *A. tumefaciens* strain GV3101 using the freeze-thaw method, as described by Weigel and Glazebrook (2006) [31]. The transformation was validated by culturing the bacteria on LB selection plates containing 25 µg/ml chloramphenicol, 50 µg/ml kanamycin, 50 µg/ml gentamicin, and 25 µg/ml rifampicin. For plant transformation, the *A. tumefaciens* culture was mixed with a dipping

solution containing 0.05% Silwet-77 and 5% sucrose. The transgenic *A. thaliana* plants were then generated via *A. tumefaciens* GV3101-mediated transformation using the floral dip method at the anthesis stage, following the protocol by Zhang et al. (2006) [32]. These first-generation (T₁) transgenic lines carrying and expressing the *STTM426* construct were screened on antibiotic selection plates supplemented with hygromycin B (50 µg/mL) and confirmed through genomic PCR analysis. The successfully transformed plants were then propagated to produce second (T₂) and third (T₃) generations, which underwent homozygosity screening and genomic PCR screening to confirm the presence of the *STTM426* construct. The homozygous T₃ transgenic plants were subsequently used for further data collection and analyses.

Plant materials, growth conditions, and light treatment

All *A. thaliana* plants, including wild type (WT) (ecotype Columbia) and *STTM426* transgenic lines, were cultivated in a growth chamber facility at The National University of Malaysia (UKM) under controlled long-day photoperiod conditions (16 h light/8 h dark), with a light intensity of 100 µmol m⁻² s⁻¹ (L₁₀₀), a temperature of 22 ± 1 °C, and relative humidity maintained at 60 ± 5%. Homozygous *STTM426*

transgenic lines (T_3 generation) and WT seedlings were initially germinated on selection plates containing full-strength Murashige and Skoog (MS) medium. For stratification, plates were wrapped in aluminium foil and incubated at 4 °C in the dark for three days. Hygromycin B (50 µg/mL) was added to the medium for selecting transgenic lines, while WT seedlings were grown on MS medium without selection. Following stratification, the plates were transferred to the growth chamber under the aforementioned controlled long-day conditions for approximately 10 days to promote germination. Once seedlings had developed at least two true leaves, they were transplanted into pots containing 200 g of peat moss. Each pot (9 cm in diameter) was initially watered with 200 mL of tap water. Thereafter, irrigation was reduced to 100 mL per pot per week to ensure consistent growth and prevent overwatering.

All plants were grown for 42 days under the same long-day controlled conditions. On day 43, *STTM426* transgenic lines and WT plants were exposed to high light stress treatment for an additional 3 days, completing a total growth period of 45 days. During this final phase, plants were exposed to a 16 h light/8 h dark photoperiod under varying high light intensities, including photosynthetic photon flux densities (PPFD) of 1,000 µmol m⁻² s⁻¹ (L_{1000}), 1,300 µmol m⁻² s⁻¹ (L_{1300}), and 1,500 µmol m⁻² s⁻¹ (L_{1500}), as well as extremely high light stress levels at 1,800 µmol m⁻² s⁻¹ (L_{1800}) and 2,000 µmol m⁻² s⁻¹ (L_{2000}). Temperature (22 ± 1 °C) and humidity (60 ± 5%) were maintained throughout. The light intensity was quantified using an LI-180 Spectrometer (Li-COR, USA) and further validated with an LI-190R Quantum Sensor and an LI-250 A Light Meter (Li-COR, USA). After 45 days of vegetative growth and light treatment exposure, the plants were subjected to downstream analyses, including the quantification of miR426 and *ELIP2* transcript levels, as well as physiological assessments of chlorophyll content and photosynthetic efficiency in both *STTM426* transgenic lines and WT plants.

RNA extraction and qRT-PCR analysis

Fresh leaf tissue from transgenic and WT *A. thaliana* plants, specifically the 10th rosette leaf counted from the base, was ground into a fine powder using liquid nitrogen and a TissueLyser LT (Qiagen, Germany). Total RNA was extracted using the PureLink® Plant RNA Reagent (Life Technologies, Netherlands) by following the manufacturer's protocol. RNA concentration and purity were assessed using a NanoDrop ND-1000 Spectrophotometer (Thermo Fisher Scientific, USA). For target transcript analysis, total RNA was treated with a genomic DNA (gDNA) remover and reverse-transcribed into complementary DNA (cDNA) using the ReverTra Ace™ qPCR RT Master Mix with

gDNA Remover (Toyobo, Japan), according to the manufacturer's protocol. For miRNA expression analysis, a stem-loop primer structure was designed with a universal backbone and specific trigger elements using the Vazyme online tool (<https://www.vazymeglobal.com/cetool/mirna.html>) (Supplementary Table S1). Total RNA was treated with gDNA remover, and cDNA synthesis was performed using the SuperScript® III First-Strand Synthesis SuperMix for quantitative real-time PCR (qRT-PCR) (Thermo Fisher Scientific, USA), following the manufacturer's protocol.

Next, the synthesized cDNA was used as a template for quantitative reverse transcription polymerase chain reaction (qRT-PCR), performed on a CFX96 Real-Time PCR system (Bio-Rad, USA). Prior to the analysis, a 20 µL PCR reaction mixture was prepared using 2× Taq Pro Universal SYBR qPCR Master Mix (Vazyme, China), following the manufacturer's protocol to determine the expression levels of *ELIP2* and miR426. To ensure precise quantification, *ACT2* was used as the reference gene for normalizing samples in conventional qRT-PCR targeting mRNA transcript, while *U6* small nuclear RNA (snRNA) served as the reference gene for stem-loop qRT-PCR in miRNA expression analysis. The qPCR reaction began with an initial denaturation at 95 °C for 3 min, followed by 40 cycles of amplification, consisting of denaturation at 95 °C for 10 s, annealing at 58 °C, and extension at 60 °C for 30 s. To confirm the absence of gDNA contamination, a melting curve analysis was performed for each sample using a temperature ramp from 65 °C to 95 °C, with 0.5 °C increments and 0.05 s intervals at each temperature step. The analysis was conducted using three independent biological replicates, each with three technical replicates. The relative expression levels were measured using the $2^{-\Delta\Delta CT}$ analysis method to compare expression between wild type and transgenic under different light conditions [33], and the primer sequences used in qRT-PCR are listed in Supplementary Table S2.

Determination of chlorophyll a, chlorophyll b, and total chlorophyll content

The absolute quantification method for chlorophyll content was adapted from Liang et al. (2017) [34], with modifications involving the use of 90% acetone (Hamburg Chemical, Germany) for extraction. All measurements were carried out specifically on the 11th rosette leaf counted from the base of *A. thaliana* plants. To ensure consistent sample normalization, four uniform leaf discs (5 mm in diameter) were punched from each leaf and homogenized in liquid nitrogen before overnight incubation in 90% acetone under dark conditions. After centrifugation at 15,000 × g for five minutes at room temperature (23 ± 2 °C), the supernatant was analysed using a UV-Vis Spectrophotometer DU-730

(Beckman Coulter, USA) at 645 nm (chlorophyll *b*) and 663 nm (chlorophyll *a*). The analysis was conducted using nine independent biological replicates, each with three technical replicates. Chlorophyll concentrations were calculated following Arnon's equations [35] as follows:

$$\text{Chlorophyll } a \text{ (mg/g)} = 12.70 (A_{663}) - 2.69 (A_{645}).$$

$$\text{Chlorophyll } b \text{ (mg/g)} = 22.90 (A_{645}) - 4.68 (A_{663}).$$

$$\text{Total chlorophyll (mg/g)} = 20.2 (A_{645}) + 8.02 (A_{663}).$$

Evaluation of photosynthetic efficiency and photoprotective mechanisms using gas exchange and chlorophyll *a* fluorescence parameters

Gas exchange and chlorophyll *a* fluorescence parameters were measured using the LI-6800 portable gas exchange system (Li-COR, USA). Prior to measurement, each plant was exposed to dark adaptation overnight. Gas exchange measurements were conducted to determine the CO₂ assimilation rate (*A*), transpiration rate (*E*), and stomatal conductance (*g_{sw}*). Chlorophyll *a* fluorescence analysis was used to evaluate the maximum quantum efficiency of photosystem II (PSII) under light-adapted conditions (F_v'/F_m'), photochemical quenching (*qP*), and non-photochemical quenching (*NPQ*), providing insights into photosynthetic efficiency and photoprotective mechanisms. All measurements were conducted specifically on the 12th rosette leaf, counted from the base of *A. thaliana* plants. The selected leaf was securely clamped in the LI-6800 Multiphase Flash Fluorometer chamber (leaf chamber gasket area: 2 cm²; diameter: 1.6 cm). Environmental conditions, including leaf temperature (22 ± 1 °C), relative humidity (60 ± 5%), vapour pressure deficit (1 kPa), light intensity (100 μmol m⁻² s⁻¹), and CO₂ concentration (400 μmol mol⁻¹), were automatically regulated to stimulate natural conditions [36]. Blue light was maintained at 10% of the total photosynthetic photon flux density (PPFD), with the remaining 90% supplied by red light, to optimize stomatal aperture. Readings were recorded once the gas exchange system stabilized, ensuring measurements were taken before the leaves responded to the new environmental conditions.

Assessment of morphological traits

A comprehensive evaluation of morphological traits was conducted once the plants reached physiological maturity, focusing on rosette diameter (RD, cm), and total biomass (TB, mg). To determine RD, all samples were visualized using a trinocular microscope (Olympus, Japan) under low magnification with reflected surface lighting to ensure uniform image capture. High-resolution images were obtained and quantitatively analysed using IMAGE J 1.52 software. Meanwhile, TB was assessed by measuring the dry weight

of the entire plant. Fresh samples were harvested, thoroughly washed to remove soil particles, and oven-dried at 70 °C for 48 h using a laboratory air-drying oven. The dry weight was then measured using an analytical balance. All phenotypic data were meticulously recorded to investigate the morphological responses to varying light intensities.

Statistical analysis

All statistical analyses were conducted using GraphPad Prism (version 10.3.1). One-way and two-way analyses of variance (ANOVA) were performed, followed by Tukey's post hoc test, with different letters indicating statistically significant differences ($P \leq 0.05$). Error bars represent the standard error of the mean (SE). For expression profiling based on qRT-PCR, data were obtained from three independent biological replicates, each with three technical replicates. In contrast, for physiological analyses including chlorophyll content, photosynthetic efficiency, and photoprotective mechanisms, data were based on nine independent biological replicates, each with three technical replicates.

Results

Identification of putative miRNA targeting *ELIP2* transcripts

This study employed the psRNATarget V2 platform, an established and robust computational tool for miRNA target prediction in plants, to identify potentially miRNAs regulating *ELIP2* transcripts. A core feature of this platform is the Expectation Score (ES), which evaluates the complementarity between miRNA and target mRNA sequences by assigning penalties for mismatches, thereby estimating the strength of interaction [26].

Out of 427 annotated *A. thaliana* miRNAs screened using default parameters, four potentially miRNAs were initially predicted to target *ELIP2* transcripts. Among them, miR426 emerged as the top-ranked miRNA candidate, exhibiting the lowest ES of 3.5, which falls within the recommended predictive confidence range of 3.0 to 5.0 (Table 1). A lower ES indicates a stronger predicted binding affinity and thus, a higher probability of regulatory interaction. In contrast, miRNAs with higher ES values (>3.5) were considered to have weaker binding potential and are more likely to be false positives, indicating a lower likelihood of functional interaction with *ELIP2*.

Table 1 List of potentially miRNAs targeting the *ELIP2* transcript in *A. thaliana* based on psrnatarget V2 analysis

miRNA	Expectation Score (ES)	miRNA and target transcript alignment
ath-miR426	3.5	21-GCAUCCUGUU UAAAGGUUUU-1::: 912-UUUAAGGACAA AUUUGGAGAA-932
ath-miR5015	4.0	21-CUUCUGAUGUG UAUUGUGGUU-1::: 185-AUUGACUACAC GCAACAUCAG-205
ath-miR447a-3p	5.0	22-GUUGUUUUGUA GAGCAGGGGUU-1 ::::: ::::: 208-ACACAAACCAA CUCUUCUCAA-229
ath-miR865-3p	5.0	21-AACCUAUUUAA ACUCCUUUUU-1 ::: ::::: 915-AAGGACAAAUU UGGAGAAUGA-935

Suppression of miR426 by STTM426 significantly enhances *ELIP2* expression under varying light conditions

STTM is a pivotal technology employed to investigate the suppressive impact of the miRNAs [37]. In *A. thaliana*, STTM primarily induces the degradation of targeted miRNAs. To evaluate its effectiveness in suppressing miR426, we designed and constructed STTM426 by

integrating the enhanced d35S promoter to constitutively target the sequence of miR426. The construct included two non-cleavable miRNA binding sites complementary to the mature miR426 sequence, with a 3-nt bulge at the 11th nucleotide position and connected by a 48-nt spacer. The resulting homozygous T₃ transgenic plants were then analysed for miR426 expression levels in comparison to WT control plants. Notably, under normal light conditions (L₁₀₀), STTM426 transgenic plants exhibited a 0.90-fold reduction in miR426 abundance compared to WT (Fig. 2a). This result confirmed the efficiency of STTM426 in silencing miR426, which corresponded with an approximately 0.90-fold increase in the expression of its target transcript, *ELIP2*, relative to WT (Fig. 2b).

The effectiveness of STTM426 in silencing miR426 and enhancing the expression of its target transcript, *ELIP2*, was further assessed under varying light intensities. Under high light stress (L_{1000–1500}), miR426 suppression was comparable to that under normal light conditions (L₁₀₀), with a 0.85- to 0.91-fold reduction compared to WT, accompanied by a significant increase to 0.60- to 0.90-fold in *ELIP2* expression. Interestingly, under extreme high light stress (L_{1800–2000}), miR426 suppression was even more pronounced, with a 0.94- to 0.97-fold reduction, which correlated with a marked increase in *ELIP2* expression to 1.50- to 1.80-fold compared to WT. These results demonstrated that miR426 suppression is correlated with increased *ELIP2* expression, with more prominent effects under extreme high light conditions.

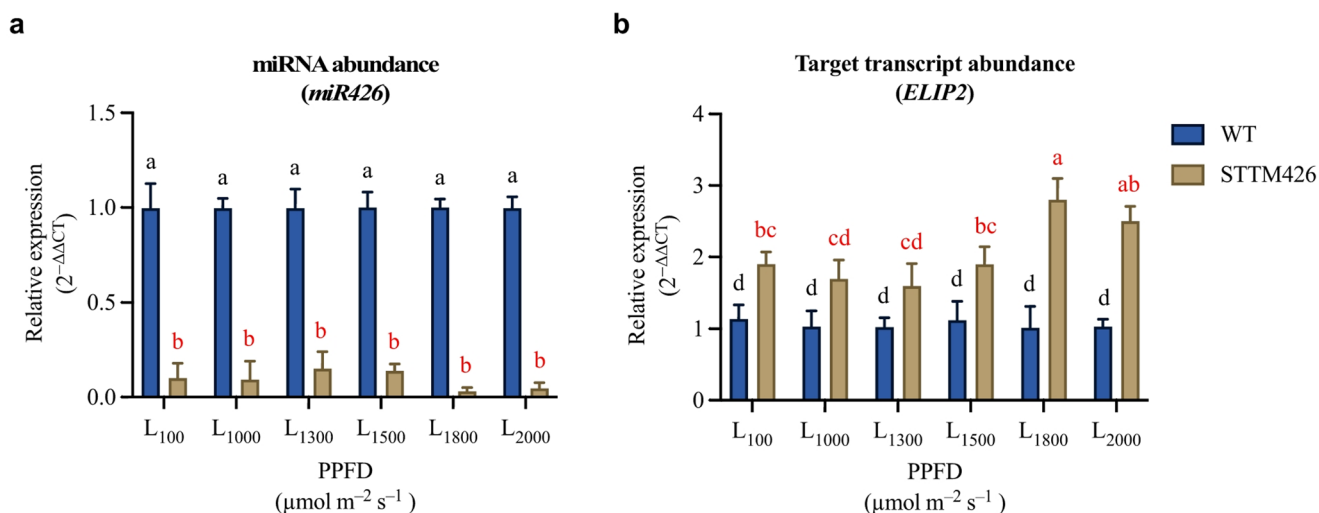


Fig. 2 Changes in miRNA and target transcript expression levels in *A. thaliana* under different light intensity treatments. **(a)** Relative expression of miR426 and **(b)** target transcript *ELIP2* in wild type (WT) and transgenic line (STTM426) across increasing photosynthetic photon flux densities (PPFD) ranging from L₁₀₀ to L₂₀₀₀, corresponding to 100 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Data represent means $\pm$ SE of three technical replicates from three independent biological experiments ($n=3$), with

error bars indicating standard errors. Statistically significant differences were determined using two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons. Different lowercase letters above the bars indicate significant differences ($P \leq 0.05$). ANOVA results are indicated by red letters for STTM426 and black letters for WT

Modulation of chlorophyll content through *ELIP2* upregulation under varying light intensities

Chlorophyll *a*, chlorophyll *b*, and total chlorophyll content were analysed in *A. thaliana* T₃ homozygous transgenic plants compared to WT plants. The absolute quantification results revealed that transgenic plants exhibited lower chlorophyll levels compared to WT plants under normal conditions (L₁₀₀). Specifically, transgenic plants contained 0.28 mg/g less total chlorophyll than WT plants. This result indicates that enhanced *ELIP2* expression under miR426 suppression reduced free chlorophyll production, resulting in a subsequent decrease of total chlorophyll content. Further analysis of chlorophyll composition showed that STTM426 transgenic plants exhibited a significant decrease in chlorophyll *a*, while chlorophyll *b* levels were elevated compared to WT plants.

Consistent with the trend observed under normal conditions, the total chlorophyll content and chlorophyll composition (chlorophyll *a*) decreased with increasing light intensity in WT plants (Fig. 3a, b). This reduction was primarily attributed to the decline in chlorophyll *a*, which constitutes most of the total chlorophyll. Under L₁₀₀₀, STTM426 plants recorded a total chlorophyll content of 2.46 mg/g and chlorophyll *a* of 1.34 mg/g, while WT plants had 2.80 mg/g and 1.77 mg/g, respectively. Similarly, at L₁₃₀₀, STTM426 plants recorded a total chlorophyll content of 2.37 mg/g and chlorophyll *a* of 1.31 mg/g, compared to WT plants which had 2.56 mg/g and 1.35 mg/g, under these high light intensity conditions. Overall, STTM426 plants exhibited a reduction in total chlorophyll content under L₁₀₀₀ to L₁₃₀₀, with differences of 0.34 mg/g and 0.19 mg/g compared to WT, respectively. Intriguingly, across all the high (L₁₅₀₀) and extremely high (L_{1800–2000}) light intensity conditions, STTM426 plants exhibited a relatively stable total chlorophyll content compared to WT.

While STTM426 plants exhibited a reduction in both chlorophyll *a* and chlorophyll *b* contents compared to WT under normal and moderately high light conditions, stabilization of chlorophyll levels was observed under extremely high light stress (Fig. 3b, c). This resulted in a significantly higher total chlorophyll content (both chlorophyll *a* and chlorophyll *b*) in STTM426 plants compared to WT plants beyond L₁₅₀₀. Under extreme high light conditions, a significant decrease in total chlorophyll content (from L₁₈₀₀ to L₂₀₀₀) and chlorophyll *a* (at L₂₀₀₀) was observed in WT plants compared to STTM426. The observed decline in WT was primarily due to a reduction in chlorophyll *b* (Supplementary Table S3). In WT plants, chlorophyll *b* content peaked at L₁₃₀₀ and decreased thereafter. At this light intensity, both plants showed similar chlorophyll *a* content, though WT plant had a slightly higher chlorophyll *b* content. The

phenotypic effects of high light stress also became apparent from L₁₀₀₀ to L₂₀₀₀, during which STTM426 plants maintained a lush green rosette compared to WT plants, despite the rosette diameter of WT plants was larger than that of STTM426 at L₁₀₀ (Figs. 3d and 7a). These results indicate that miR426 suppression enhances STTM426 plant adaptation to high light stress by stabilizing chlorophyll content. This supports the proposed role of *ELIP2* in preserving photosynthetic pigments and enhancing resistance to chlorophyll degradation.

Photosynthetic efficiency under varying light intensities

The changes in photosynthetic efficiency in this study were assessed using key physiological parameters such as CO₂ assimilation rate (*A*), transpiration rate (*E*), and stomatal conductance rate (*g_{sw}*). The statistical analysis revealed a significance response pattern in CO₂ assimilation rate between transgenic and WT plants across varying light intensities (Fig. 4a). Under normal light conditions (L₁₀₀), transgenic plants exhibited a lower CO₂ assimilation rate than WT, with a 0.42 μmol m⁻² s⁻¹ reduction. Similarly, both transpiration rate and stomatal conductance in transgenic plants were lower compared to WT (Fig. 4b, c). In line with the trends observed in total chlorophyll content, these findings suggest that plants enhance photosynthetic performance under high light intensity (L_{1000–1300}) through adaptive mechanisms, including increased transpiration and stomatal conductance. These adaptations promote efficient gas exchange processes, such as CO₂ uptake and O₂ release, to support photosynthetic activity.

Furthermore, consistent with the proposed role of *ELIP2* as chlorophyll sensors in suppressing chlorophyll biosynthesis, preventing chlorophyll bleaching, and maximizing photosynthetic efficiency [21], this study found a sharp decline in CO₂ assimilation rate in WT plants under high (L₁₅₀₀) and extremely high (L_{1800–2000}) light-intensity stress conditions, compared to STTM426. Particularly, CO₂ assimilation rate in WT plants dropped by 71.9% to 93.5% at L₁₅₀₀ to L₂₀₀₀ (Supplementary Table S4), compared to STTM426 plants, highlighting the enhanced photoprotective mechanism of STTM426 through more effective activation of *ELIP2*. Additionally, WT plants showed a significant decline in both transpiration and stomatal conductance compared to STTM426. This finding demonstrates that miR426 suppression improves the ability of STTM426 plants to more effectively mitigate photooxidative stress and photoinhibition, enabling them to maintain higher photosynthetic efficiency under extreme light conditions.

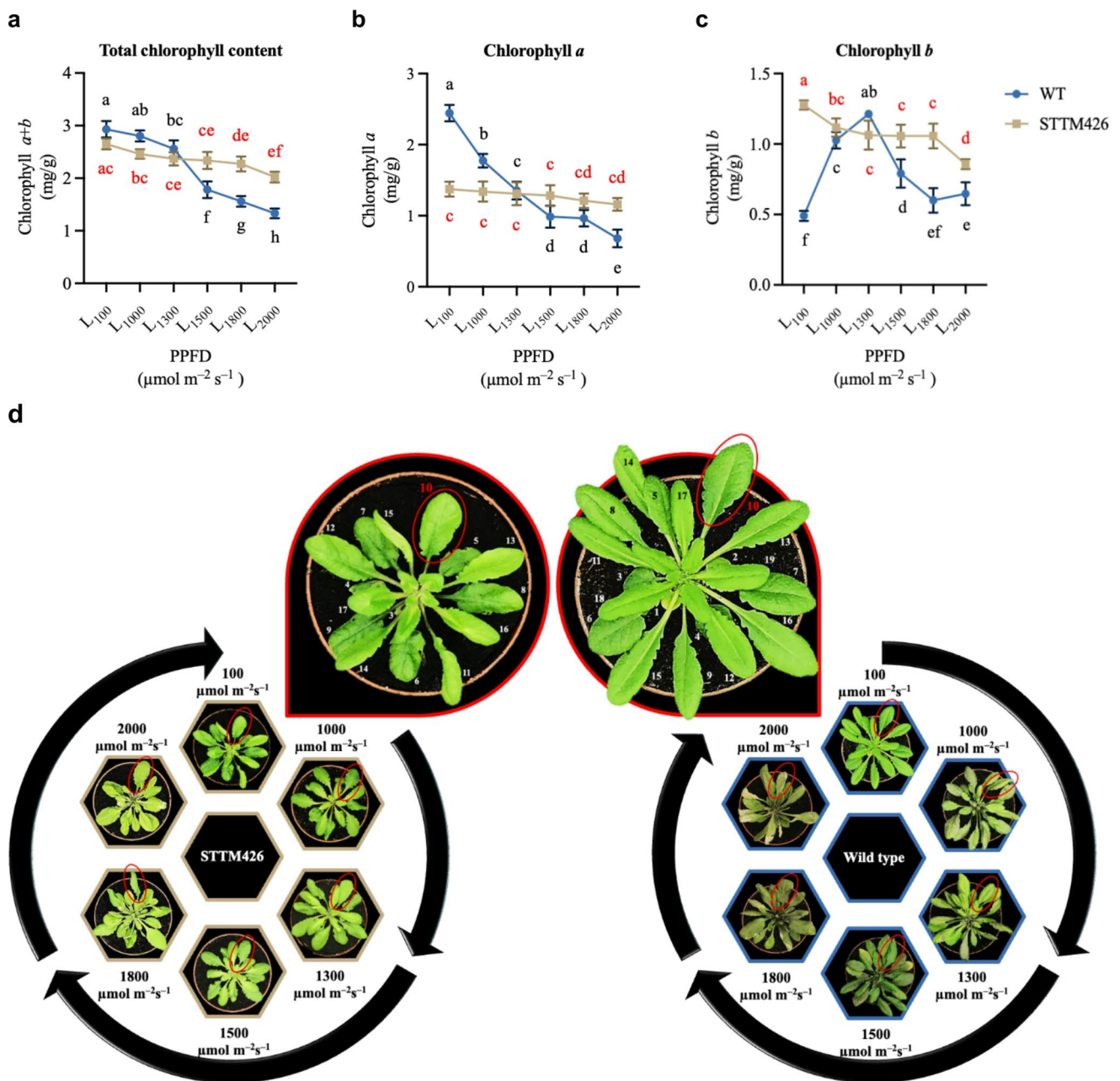


Fig. 3 Changes in total chlorophyll content and chlorophyll composition in *A. thaliana* under varying light intensity treatments. Concentrations of (a) total chlorophyll content (chlorophyll *a* + *b*), (b) chlorophyll *a*, and (c) chlorophyll *b* in the leaves of wild type (WT) and transgenic line (STTM426) across photosynthetic photon flux densities (PPFD) ranging from L₁₀₀ to L₂₀₀₀, corresponding to 100 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$. (d) Morphological changes in the rosette stage between WT and STTM426 plants under different light intensity treatments, with red circles and corresponding numbers indicating the specific

samples used for this analysis. Data represent means $\pm$ SE of three technical replicates from nine independent biological experiments ($n=9$), with error bars indicating standard errors. Statistically significant differences were determined using two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons. Different lower-case letters above the bars indicate significant differences ($P \leq 0.05$). ANOVA results are indicated by red letters for STTM426 and black letters for WT

Chlorophyll *a* fluorescence under varying light intensities

To assess the impact of varying light intensities on photoprotective mechanisms, the chlorophyll *a* fluorescence module

of the LI-6800 system was used. This measurement included key parameters such as the maximum quantum efficiency of PSII under light-adapted conditions (F_v'/F_m'), photochemical quenching (qP), and non-photochemical quenching (NPQ), providing a comprehensive understanding of leaf

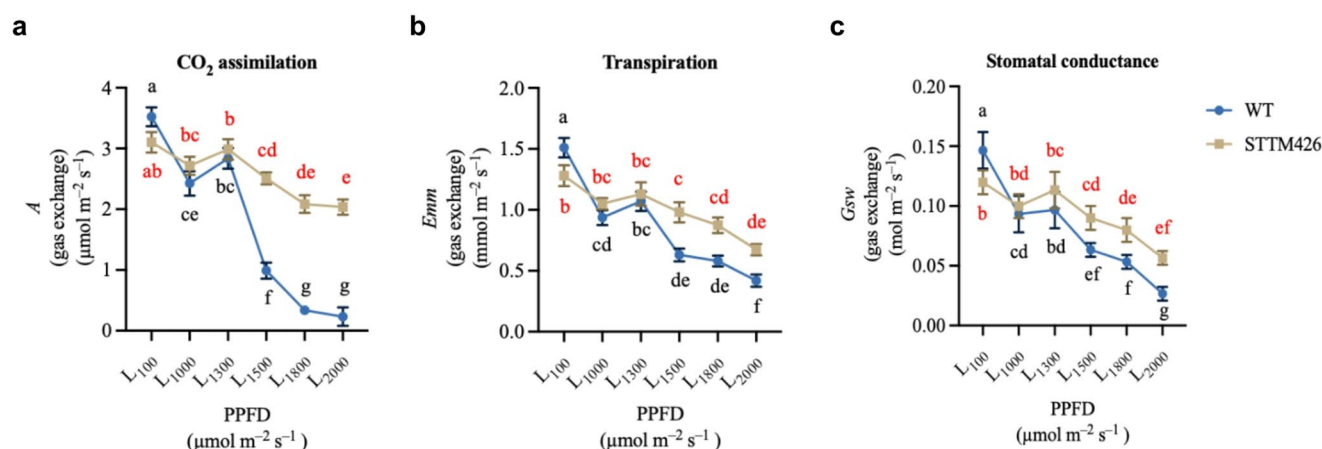


Fig. 4 Changes in photosynthetic efficiency in *A. thaliana* under varying light intensity treatments. **(a)** CO₂ assimilation rate (*A*), **(b)** transpiration rate (*E*), and **(c)** stomatal conductance rate (*g_{sw}*) changes in the leaves of wild type (WT) and transgenic line (STTM426) across photosynthetic photon flux densities (PPFD) ranging from L₁₀₀ to L₂₀₀₀, corresponding to 100 to 2000 μmol m⁻² s⁻¹. Data represent means ± SE of three technical replicates from nine independent biological experi-

ments (*n*=9), with error bars indicating standard errors. Statistically significant differences were determined using two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons. Different lowercase letters above the bars indicate significant differences (*P* ≤ 0.05). ANOVA results are indicated by red letters for STTM426 and black letters for WT

adaptation to varying light intensities. F_v'/F_m' values exhibited varying degrees of differences between transgenic and WT plants under different light intensities (Fig. 5a). Under normal light conditions (L₁₀₀), the values were nearly identical, with only a slight difference of 0.06 (7.3%) between transgenic and WT plants. The absence of a significant difference at this light intensity suggests that increased *ELIP2* expression does not negatively affect PSII efficiency under standard light conditions.

Under high light intensity stress (L₁₀₀₀), the F_v'/F_m' value in transgenic plants declined by 7.9%, compared to the 34.2% reduction in the WT. Interestingly, under more intense stress conditions of L₁₅₀₀, L₁₈₀₀, and L₂₀₀₀, although F_v'/F_m' values showed a significantly decline trend in both transgenic and WT plants, transgenic plants maintained a higher F_v'/F_m' levels compared to WT. Under extremely high light intensity (L₁₈₀₀ and L₂₀₀₀), transgenic plants exhibited a milder decline of 39.5% to 43.4%, whereas WT plants showed a steep decline of 68.3% to 74.4% (Fig. 5a, Supplementary Table S5). The significantly higher F_v'/F_m' values in transgenic plants suggest an enhanced photoprotective capacity, potentially mitigating photoinhibition and oxidative damage.

On the other hand, transgenic plants initially exhibited higher *NPQ* values at L₁₀₀ than WT plants (Fig. 5b), indicating a more active energy dissipation mechanism under normal conditions. However, at L₁₀₀₀, *NPQ* levels declined in both plant types, reaching 0.16 in WT and 0.15 in STTM426, corresponding to reductions of approximately 33.3% and 60.5% compared to their initial values at L₁₀₀ (Supplementary Table S5). Beyond L₁₃₀₀, both lines

observed a consistent decline trend in *NPQ* values, with STTM426 plants maintaining lower *NPQ* levels than WT across all subsequent light intensities. Although these differences were not statistically significant at each PPFD level, the consistently lower trend of *NPQ* in STTM426 indicates a potentially more efficient energy management strategy under high-light conditions. This trend of declining *NPQ* levels continued under high-light (L₁₃₀₀ and L₁₅₀₀) and extreme high-light stress (L₁₈₀₀), with STTM426 transgenic plants showing a more pronounced reduction than WT, although the differences were not statistically significant (Fig. 5b). Surprisingly, under L₂₀₀₀, transgenic plants exhibited a more drastic and statistically significant decrease in *NPQ*, up to 86.8% (0.05), compared to a 54.2% reduction (0.11) in WT plants (Supplementary Table S5). Visible symptoms of high light stress began to emerge from L₁₀₀₀ to L₂₀₀₀, during which STTM426 plants sustained healthy green foliage, in contrast to WT plants, which exhibited evident signs of stress (Fig. 5d).

Meanwhile, the *qP* parameter analysis measured the fraction of PSII reaction centres that remained open and active in photochemistry. A high *qP* value suggests efficient PSII function and optimal photosynthetic performance, while a low *qP* value indicates many PSII centres are closed or inactive, potentially due to stress or damage disrupting photosynthesis [38]. The observed *qP* pattern aligned with the changes in F_v'/F_m' values. At L₁₀₀, *qP* values were identical in both plants, indicating similar initial photochemical efficiency. However, as light intensity increased to L₁₀₀₀ and L₁₃₀₀, WT plants exhibited a 7.3% to 18.3% reduction in *qP*, while transgenic plants showed a 6.1% increase at

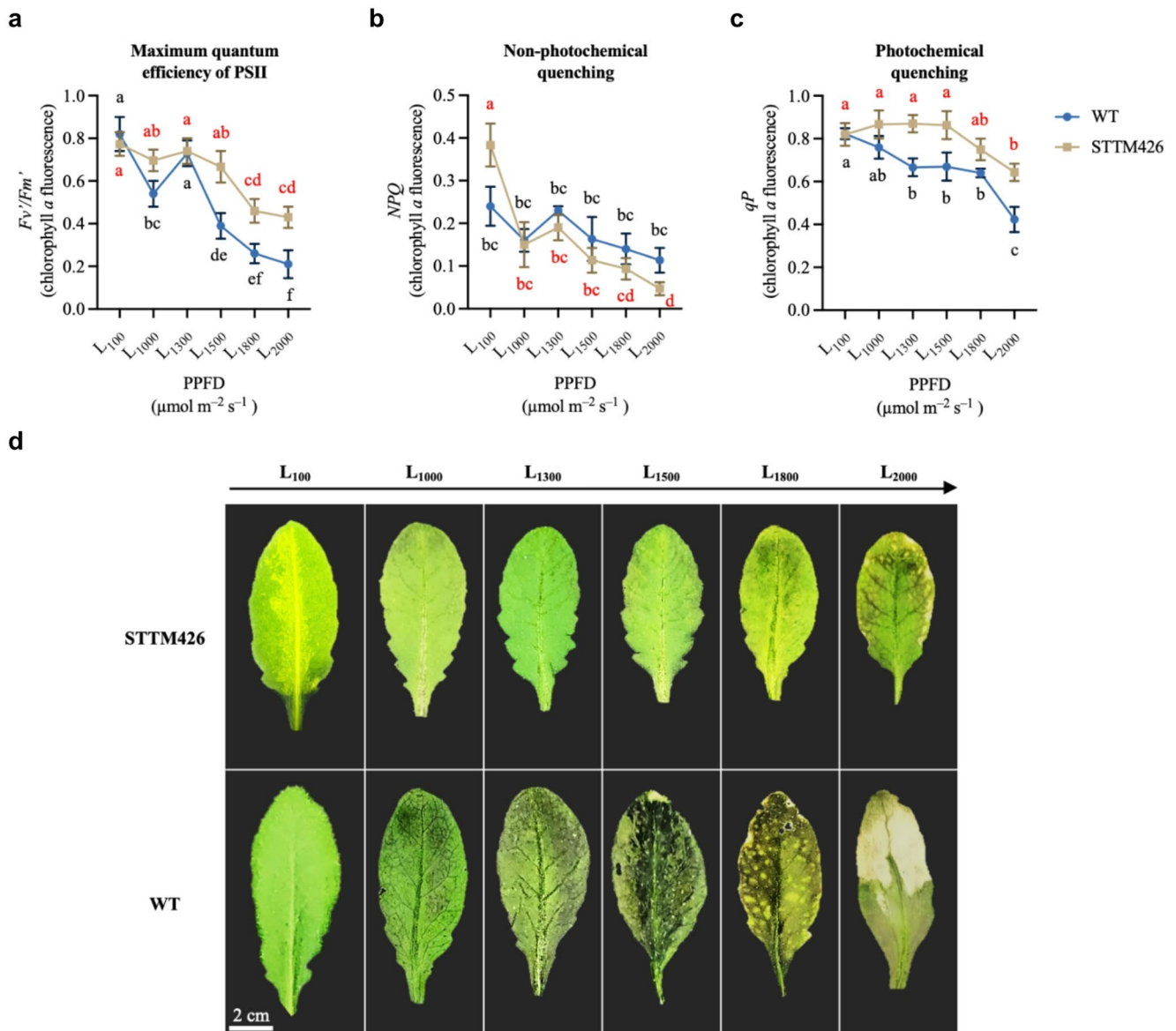


Fig. 5 Changes in photoprotective mechanisms (chlorophyll *a* fluorescence) in *A. thaliana* under varying light intensity treatments. **(a)** Maximum quantum efficiency of PSII (F_v'/F_m'), **(b)** photochemical quenching (q_P), and **(c)** non-photochemical quenching (NPQ) changes in the leaves of wild type (WT) and transgenic line (STTM426) across photosynthetic photon flux densities (PPFD) ranging from L₁₀₀ to L₂₀₀₀, corresponding to 100 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$. **(d)** Morphological changes in leaf appearance between WT and STTM426 plants under

different light intensity treatments. Data represent means $\pm$ SE of three technical replicates from nine independent biological experiments ($n=9$), with error bars indicating standard errors. Statistically significant differences were determined using two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons. Different lower-case letters above the bars indicate significant differences ($P \leq 0.05$). ANOVA results are indicated by red letters for STTM426 and black letters for WT

both intensities (Supplementary Table S5). q_P values of transgenic plants were consistently higher than WT beyond normal light conditions and remained stable before a significant decline at L₁₈₀₀ and above. Meanwhile, WT plants showed significant reductions at L₁₀₀₀ and L₁₃₀₀, followed by a plateau, before a drastic decline at L₂₀₀₀ (Fig. 5c). The ability of transgenic plants to sustain higher q_P values at high light intensities suggests an enhanced photoprotective

mechanism that minimizes excessive energy dissipation while preserving photochemical activity.

Impact of *ELIP2* upregulation on morphological responses under varying light intensities

Physiological changes, particularly variations in chlorophyll content and photosynthetic rate, can directly influence the key plant morphological traits such as rosette

diameter (RD) and total biomass (TB). Under normal light conditions (L_{100}), transgenic plants exhibited significantly lower RD and TB compared to WT plants, with approximate reductions of 2.4 cm and 118 mg, respectively. These reductions suggest an adaptive reallocation of resources by transgenic plants toward constitutive photoprotective mechanisms, likely triggered by miR426 suppression and subsequent *ELIP2* upregulation. As light intensity increased ($L_{1000-1500}$), transgenic plants continued to display reduced RD and TB, ranging from 2.2 to 2.4 cm and 75 to 85 mg, respectively (Fig. 6a, b). However, these reductions were not significantly different across light intensities within the same genotype, indicating a relatively stable stress response. In contrast, significant differences were observed when comparing transgenic and WT plants under the same light conditions, highlighting a genotype-dependent morphological adaptation to light stress. Under extremely high light stress ($L_{1800-2000}$), transgenic plants consistently maintained smaller RD values, 2.60 cm at L_{1800} and 2.20 cm at L_{2000} , although the degree of difference compared to WT plants began to plateau (Figs. 3d and 6a). Despite consistently lower TB, the transgenic lines demonstrated greater morphological stability, suggesting enhanced structural adaptation under photooxidative stress. This finding reinforces the role of *ELIP2* in modulating light stress responses by maintaining morphological integrity, mitigating oxidative damage, and stabilizing TB accumulation under extreme conditions.

Discussion

This study demonstrates that the constitutive suppression of miR426 by STTM426 in *A. thaliana* results in a sustained increase in *ELIP2* expression. The findings suggest that miR426 is likely to play a regulatory role in maintaining low *ELIP2* transcript levels under normal conditions. By suppressing miR426 through STTM426, its repressive effect on *ELIP2* is alleviated, leading to constitutively elevated *ELIP2* expression. These results align with Zhang et al. (2017), which suggest that miRNA suppression via STTM technology primarily induces a LoF effect rather than direct degradation, ultimately enhancing target transcript expression [39]. This is further supported by the observed miR426 expression patterns, which remained relatively stable in WT plants across varying light conditions but were significantly downregulated in STTM426 plants, indicating successful suppression and confirming the inverse relationship between miR426 and *ELIP2* expression. This study confirms that the *ELIP2* transcripts accumulate at a low level under normal conditions and increase under high-light stress conditions. This finding corroborates Heddad et al. (2006), which reported that *ELIP2* protein accumulation in WT plant leaves remains low, ranging from 7% to 13% of its maximum level when exposed to $1000 \mu\text{mol m}^{-2} \text{s}^{-1}$ light intensities [22]. However, Lohscheider et al. (2015) indicated that *ELIP2* transcript levels rapidly decline during the recovery phase [40]. When light intensity increases to $1500\text{--}2000 \mu\text{mol m}^{-2} \text{s}^{-1}$, *ELIP2* accumulation rises sharply, reaching 92% to 100% of its maximum level [22]. This accumulation occurs when at least 40% of PSII reaction centres are damaged, suggesting that *ELIP2* functions as a protective mechanism against extreme light stress [41]. The significant increase in *ELIP2* expression across

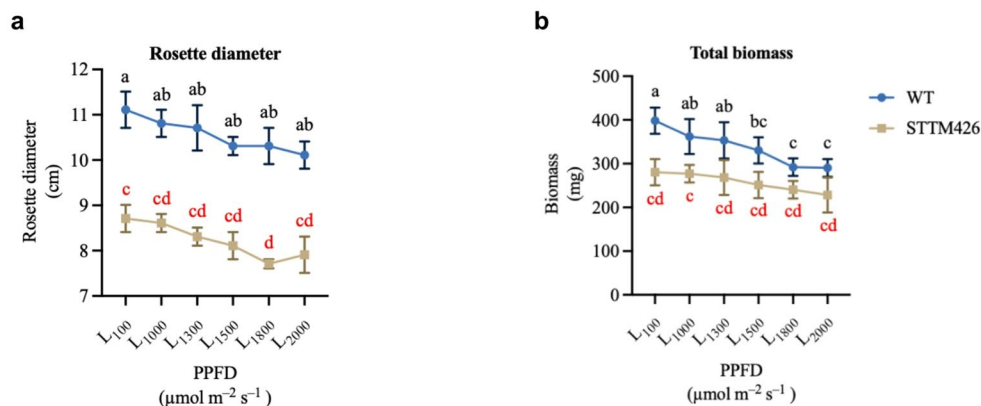


Fig. 6 Morphological changes in rosette diameter (RD) and total biomass (TB) of *A. thaliana* under varying light intensity treatments. (a) RD and (b) TB differences between wild type (WT) and transgenic line (STTM426) plants under different photosynthetic photon flux densities (PPFD) treatment ranging from L_{100} to L_{2000} , corresponding to 100 to $2000 \mu\text{mol m}^{-2} \text{s}^{-1}$. Data represent means $\pm$ SE of three technical repli-

cates from nine independent biological experiments ($n=9$), with error bars indicating standard errors. Statistically significant differences were determined using two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons. Different lowercase letters above the bars indicate significant differences ($P \leq 0.05$). ANOVA results are indicated by red letters for STTM426 and black letters for WT

various light-intensity stress treatments further indicates that the functional suppression of miR426 directly contributes to the elevated *ELIP2* expression. The upregulation of *ELIP2* plays a crucial role in the protective mechanisms against photooxidative stress and photoinhibition [42].

On the other hand, the constitutive accumulation of *ELIP2* protein in *A. thaliana* leaves noticeably decreased the total chlorophyll content of the chloroplast without significantly affecting the composition, organization, or functioning of the photosystems [21, 41]. This protective function works by minimizing damage to photosynthetic components by suppressing chlorophyllide *a*, the hydrophilic moiety of free chlorophyll, thereby preserving photosynthetic efficiency under high light-intensity stress conditions. These findings suggest that the effective suppression of miR426 by STTM426 enhances *ELIP2* expression, reducing free chlorophyll production and decreasing total chlorophyll content. The observed reduction in chlorophyll *a* level in transgenic plants, alongside the relative stability of chlorophyll *b*, suggests that *ELIP2* may preferentially regulate chlorophyll *a* due to its higher turnover rate and greater susceptibility to degradation under photoprotective mechanisms. The sustained chlorophyll *b* levels indicate that this accessory pigment in light-harvesting complexes remains functional, potentially compensating for the decline in chlorophyll *a* to maintain photosynthetic efficiency [43].

This study further elucidates the functional role of *ELIP2* in responding to high light stress by demonstrating that transgenic plants exhibit a more effective adaptive strategy for mitigating excessive light exposure. Although transgenic plants exhibit a slight reduction in total chlorophyll content and chlorophyll composition under both normal and high light conditions, they retain and stabilize chlorophyll more effectively under extreme light stress. These findings are consistent with previous research, suggesting that elevated *ELIP2* expression enhances plant resilience by regulating chlorophyll content and metabolism [8]. Notably, STTM426 transgenic plants show greater resistance to chlorophyll degradation or bleaching than WT under high-light stress. This resilience is attributed to *ELIP2*, which not only protects chloroplast components from light-induced damage but may also serve as a temporary reservoir for residual chlorophyll, preventing its complete degradation [44]. These findings further support the role of *ELIP2* in photoprotection by modulating chlorophyll homeostasis to minimize oxidative stress while maintaining functional photosynthetic complexes.

Impaired chlorophyll stabilization influences the efficiency of photosynthetic performance and the photoprotective role in *ELIP2*-accumulating plants. To assess the impact of total chlorophyll changes, further studies have been conducted to examine their effects on photosynthetic

efficiency and photoprotective mechanisms. Light intensity plays a crucial role in influencing the CO₂ assimilation rate, which reflects photosynthetic efficiency, as well as affecting transpiration rate and stomatal conductance in leaves [45]. This current study highlights that transgenic plants exhibit distinct photosynthetic efficiency patterns compared to WT plants, with lower efficiency under non-stress conditions but enhanced resilience under extreme high-light stress. This resilience is associated to increased transpiration and stomatal conductance, which help maintain cellular homeostasis and dissipate excess energy [42]. The ability of transgenic plants to sustain photosynthetic function despite reduced chlorophyll content indicates that this protective strategy not only minimizes damage to the photosynthetic apparatus but also optimizes energy utilization under high-light conditions.

Stabilizing chlorophyll in transgenic plants is essential for reducing photodamage by optimizing light energy absorption and utilization under stress conditions. Simultaneously, photoprotective mechanisms help preventing oxidative damage caused by excessive light exposure [46]. The relationship between light intensity and F_v'/F_m' values reveals that transgenic plants sustain higher PSII efficiency than WT under increasing stress, suggesting that miR426 suppression enhances photoprotection. This effectiveness is attributed to the role of *ELIP* proteins, along with stress enhanced proteins (*SEP*) and one helix proteins (*OHP*), in stabilizing PSII. These proteins contribute to reinforcing the PSII repair cycle, indirectly supporting chlorophyll biosynthesis and pigment stability, particularly under high light stress conditions [44]. Additionally, the activation of *ELIP2* is influenced by the level of stress experienced by the plant, with its expression becoming active in photoprotective mechanisms only when triggered by stress signals [47]. A previous study reported that the response of WT plants to high light-intensity stress (L_{1000}) exhibited a sharp decline in F_v'/F_m' values (also related to F_v'/F_m'), accompanied by a significant increase in lipid peroxidation levels [48]. This condition caused PSII inactivation and damage to the photosynthetic membrane due to photooxidative stress.

The *NPQ* parameter measures a plant's ability to dissipate excess light energy as heat, protecting the photosynthetic system from damage caused by excessive light exposure [49]. A high *NPQ* value reflects an effective protective mechanism that enables the plant to manage and dissipate surplus energy, preventing photosynthetic damage efficiently. Meanwhile, a low *NPQ* value indicates a reduced capacity to cope with light stress, increasing the risk of photodamage [50]. In this study, unlike WT plants that rely heavily on *NPQ* to dissipate excess energy as heat, STTM426 transgenic plants appeared to adopt an alternative strategy through elevated *ELIP2* expression, although

the differences were not statistically significant. We propose that elevated *ELIP2* expression in STTM426 plants may contribute to photoprotection by stabilizing chlorophyll and preventing excessive accumulation under high-light stress, thereby reducing the reliance on *NPQ*-mediated dissipation. While *NPQ* rapidly dissipates excess light energy as heat, its reliance on proton gradients and slow recovery may limit plant growth under high or fluctuating light conditions [51]–[52]. In contrast, *ELIP2*-mediated photoprotection offers a more stable and efficient alternative. A previous study suggested that the suppression of chlorophyll biosynthesis, alongside increased *ELIP2* expression, indicates that *ELIP2* function is not directly linked to *NPQ*, which is typically associated with high-light stress protection [53]. This notion is further supported by Rossini et al. (2006) [48], which examined *elip1* and *elip2* mutations in comparison to WT plants, challenging the hypothesis proposed by Montané & Kloppstech (2000) [54]. Their findings demonstrated that *ELIP* proteins do not function as antennae for dissipating excess light energy as heat but instead play a crucial role in chlorophyll protection. Although the difference in *NPQ* values between transgenic and WT plants is not statistically significant, the lower *NPQ* levels observed in transgenic plants suggest reduced photooxidative stress and photoinhibition, reinforcing their enhanced photoprotective capacity.

The inverse correlation between increased *qP* and decreased *NPQ* in transgenic plants compared to WT suggests a finely tuned photosynthetic protective mechanism that effectively utilizes *NPQ* without compromising photochemical efficiency. To support these findings, a previous study on *Vitis vinifera* L. (grapevine) leaves demonstrated a similar correlation pattern, where *qP* values exhibited a decline followed by stabilization, while *NPQ* values showed an increasing and stable trend as light intensity stress increased [55]. These adaptations allow transgenic plants to sustain CO₂ assimilation and overall photosynthetic performance under extreme light conditions, reinforcing the role of *ELIP2* in enhancing photoprotection and photosynthetic resilience. This is further supported by the observed morphological changes, which serve as indicators of photoprotective mechanisms. Transgenic STTM426 plants consistently exhibited smaller RD and reduced TB compared to WT plants across all light intensity conditions. These reductions became more pronounced with increasing light intensity, reflecting an adaptive strategy by transgenic plants to limit exposure to high-light stress. Interestingly, although STTM426 plants maintained lower RD and TB than WT, the decline in these traits plateaued under extremely high light, suggesting a degree of morphological stability. Although biomass reduction may appear disadvantageous, it serves as an adaptive mechanism to alleviate metabolic stress and mitigate photooxidative damage in the long term [56].

Collectively, the upregulation of *ELIP2* through miR426 suppression by STTM426 not only triggers molecular and physiological responses but also drives morphological adaptations, contributing to a more integrated and efficient strategy for coping with extreme high-light stress conditions. Beyond confirming miR426 regulatory effect, these findings suggest that *ELIP2* functions as a dual role modulator, including stabilizing the photosynthetic machinery and photoprotective mechanisms through chlorophyll regulation while concurrently influencing growth-related morphological traits (Fig. 7). Although suppression of miR426 by STTM426 significantly upregulated *ELIP2* transcripts and improved chlorophyll stability, the coordination towards *ELIP2* proteins level involved in chlorophyll biosynthesis particularly within the TBP may influence chlorophyll production and stability remains unclear.

To investigate this, we initially included the *elip2* mutant in our experimental procedure under both normal (100 $\mu\text{mol m}^{-2} \text{s}^{-1}$) and high-light (1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$) conditions. However, the *elip2* mutant plants rapidly exhibited severe stress symptoms including wilting, chlorosis, and necrosis, particularly following exposure to high light, which prevented the acquisition of reliable physiological and molecular data. These limitations restricted the inclusion of *elip2* mutant data under higher stress conditions. This observation is consistent with the previous reports that highlight the critical role of *ELIP2* response to photooxidative stress and photoinhibition particularly when photosystem II (PSII) damage increases by 40% under extreme light intensities [22]. Consequently, the assessment of *elip2* mutant under only moderate light conditions is insufficient to fully validate its protective function.

In addition to miR426, our in-silico analysis also predicted other candidate miRNAs such as miR5015, miR447a-3p, and miR865-3p as potential regulators of *ELIP2*. Although these interactions were not experimentally validated in this study, their possible involvement suggests that *ELIP2* regulation is likely multifactorial and controlled by a network of miRNAs. Furthermore, direct ROS quantification methods such as nitroblue tetrazolium (NBT) and 3,3'-diaminobenzidine (DAB) histochemical staining would provide valuable complementary evidence to strengthen our conclusions regarding the photoprotective mechanisms in STTM426 plants. To overcome these constraints and deepen our understanding of *ELIP2* function, future studies should incorporate RNA ligase-mediated rapid amplification of cDNA ends (RLM-RACE) or degradome sequencing to experimentally confirm not only direct miR426-mediated cleavage of *ELIP2* but also validate other predicted miRNA–*ELIP2* interactions. Additionally, integration of transcriptomic, proteomic, and metabolomic approaches will be essential to uncover the protein-level

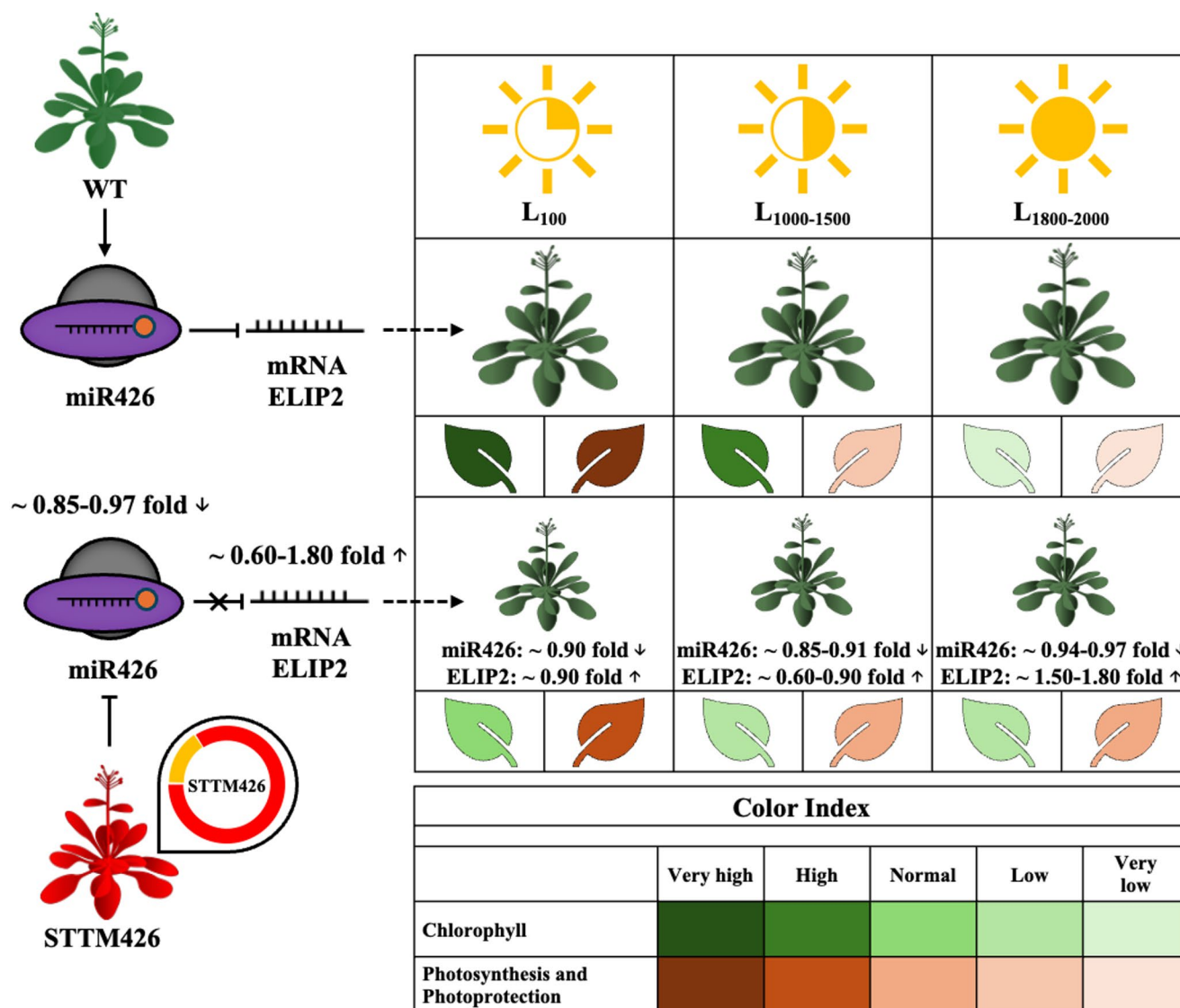


Fig. 7 Illustrative summary model showing the integrated molecular, physiological, and morphological responses of transgenic STTM426 plants compared to wild type (WT) under varying light intensities. The model highlights the regulatory role of miR426 suppression in enhanc-

ing *ELIP2* expression, leading to improved chlorophyll stability, photosynthetic performance, photoprotective mechanisms, and adaptive morphological changes in response to light stress

interactions, enzymatic activity, and broader regulatory networks through which *ELIP2* modulates chlorophyll biosynthesis and photoprotection.

Conclusion

These preliminary findings highlight miR426 as a key post-transcriptional regulator of *ELIP2*, offering critical insight into plant adaptation under excessive light stress. Suppressing miR426 via STTM426 was associated with a significant upregulation of *ELIP2* expression in transgenic *A. thaliana*, which was suggestively linked to improved photoprotection, chlorophyll retention, and photosynthetic efficiency

under extreme light conditions. Transgenic plants exhibited trends of enhanced photosynthetic performance including increased CO_2 assimilation, stomatal conductance, and transpiration rates, as well as improved photoprotective responses, reflected by the elevated Fv'/Fm' , NPQ , and qP values. Additionally, these physiological advantages were accompanied by morphological adaptations that minimized light-induced stress. Collectively, our results provide predictive evidence that *ELIP2* may serve a dual role in maintaining chlorophyll homeostasis and supporting morphological stability under high light conditions through miR426 suppression. While further validation is required, this work highlights *ELIP2* as a potential regulatory node in photoprotective adaptation and a candidate pathway for

improving photosynthetic performance and agricultural resilience under high light conditions.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval Not applicable.

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